

While a cell is being executed, no other cell can be executed. If you try to execute another cell, its execution will not start until the first cell has finished its execution. Once the execution is finished, the header of the cell will be replaced by the next number of execution. Since this will be the first cell executed, the number shown will be 1. If the process of importing the libraries is correct, no output cell is produced.

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
```

For simplicity, other chapters in this book will avoid writing these imports.

The DataFrame Data Structure

The key data structure in Pandas is the DataFrame object. A DataFrame is basically a tabular data structure, with rows and columns. Rows have a specific index to access them, which can be any name or value. In Pandas, the columns are called Series, a special type of data, which in essence consists of a list of several values, where each value has an index. Therefore, the DataFrame data structure can be seen as a spreadsheet, but it is much more flexible. To understand how it works, let us see how to create a DataFrame from a common Python dictionary of lists. First, we will

create a new cell by clicking **Insert** **Insert Cell Below** or pressing the keys **Ctrl +** . Then, we write in the following code:

```
data = {'year': [
    2010, 2011, 2012,
    2010, 2011, 2012,
    2010, 2011, 2012
],
'team': [
    'FCBarcelona', 'FCBarcelona',
    'FCBarcelona', 'RMadrid',
    'RMadrid', 'RMadrid',
    'ValenciaCF', 'ValenciaCF',
    'ValenciaCF'
],
'wins': [30, 28, 32, 29, 32, 26, 21, 17, 19],
'draws': [6, 7, 4, 5, 4, 7, 8, 10, 8],
'losses': [2, 3, 2, 4, 2, 5, 9, 11, 11]
}
football = pd.DataFrame(data, columns = [
    'year', 'team', 'wins', 'draws', 'losses'
])
```

In this example, we use the pandas DataFrame object constructor with a dictionary of lists as argument. The value of each entry in the dictionary is the name of the column, and the lists are their values.

The DataFrame columns can be arranged at construction time by entering a key-word columns with a list of the names of the columns ordered as we want. If the